

Lecture 3: Discrete Random Variables, Expectation

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- 1 Discrete random variables::: {.callout-note icon=false}## Definition: Random variable / PMFA random variable X is a function from Ω to $\mathbb{R}$. It is discrete if it takes finitely or countably many values, with PMF $p_X(x) = P(X = x)$, $\sum_x p_X(x) = 1$::: . . .::: {.callout-note icon=false}## Definition: Expectation

$$E[X] = \sum_x x p_X(x),$$

the probability-weighted average, i.e. the long-run average value of X :::—::: {.callout-warning icon=false}## FactBernoulli(p): $p_X(1) = p$, $E[X] = p$. Binomial(n, p): $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$, $E[X] = np$:::## PMFs in practice 1

—title: “Lecture 3: Discrete Random Variables, Expectation”author: “Jamie Haddock”format: revealjs: output-file: Lecture3_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
from math import combn, p = 20, 0.3samples = rng.binomial(n, p, size=100000)values, counts = np.unique(samples, return_counts=True)
```

—# Case Study: The Randomized Kaczmarz MethodGiven $Ax = b$ with rows $a_1, \dots, a_m$, randomized Kaczmarz repeatedly picks a random row index I and projects onto $\{x : a_I^\top x = b_I\}$::: {.callout-note

icon=false}### Definition: Row-sampling distributionThe row I is sampled with

$$p_i = \frac{\|a_i\|^2}{\|A\|_F^2}, \quad i = 1, \dots, m,$$

where $\|A\|_F^2 = \sum_i \|a_i\|^2$ – a genuine PMF over $\{1, \dots, m\}$, built from the data in A .: . .

```
m, n = 8, 4A = rng.normal(size=(m, n))x_true = rng.normal(size=n)b = A @ x_truerow_norms_sq = np.sum(A**2, axis=0)x = np.zeros(n)errors = []for k in range(200):    i = rng.choice(m, p=p)    a_i, b_i = A[i], b[i]    x = x + a_i * b_i / row_norms_sq[i]
```

We'll return to Kaczmarz twice more: once for convergence in probability (Week 8), and once for Markov chains (Week 10).### Reading & HomeworkReading: §2.1-2.4 (*Bertsekas & Tsitsiklis, Introduction to Probability*).**Due today:** Homework 1 (Gradescope).